

PV–PORT 135 A port board

Rev F0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Ports A (PV) and B (bus/battery) of PV–P75, common negative BUS–

01 Connectors and sensor supply

PORT 26–way to CTRL–C2000, 14–way to SYS–IO–AUX, ID + hold readback, +24V → +3V3S buck, AGND tie, PE bond

File: sheets/01_io.kicad_sch

02 Port A power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/02_portA_power.kicad_sch

03 Port A sensing and hold logic

VA bus side, VAX terminal side (AMC3330), IA current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/03_portA_sense.kicad_sch

04 Port B power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/04_portB_power.kicad_sch

05 Port B sensing and hold logic

VB bus side, VBX terminal side (AMC3330), IB current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/05_portB_sense.kicad_sch

06 Insulation measurement

AUX1 = V(PE–BUS–), AUX2 = V(PE–A+) (AMC3330); 992 k test strings A+ → PE, BUS– → PE, each with a common–source SiC pair (TPSI3050)

File: sheets/06_imd.kicad_sch

Common negative: port A and port B share BUS– (non–isolated buck–boost), so both ports are one HV domain; the main contactors switch the + pole only, fuses are in both poles. Chassis–mounted parts (terminals, fuses, CM choke, contactors, RST 200 resistors, thermostats, shunt) are drawn here with their real MPN or an RFQ / CUSTOM specification.

Isolators (checked on every build): AMC3330/AMC3302, TPSI3050–Q1, CNY65B, contactor and relay coil–contact. Declared HV–PE crossings: MOV3 of each varistor Y, Y capacitors, the PE–side SiC of each insulation–measurement pair, the PE–end element of the AUX dividers.

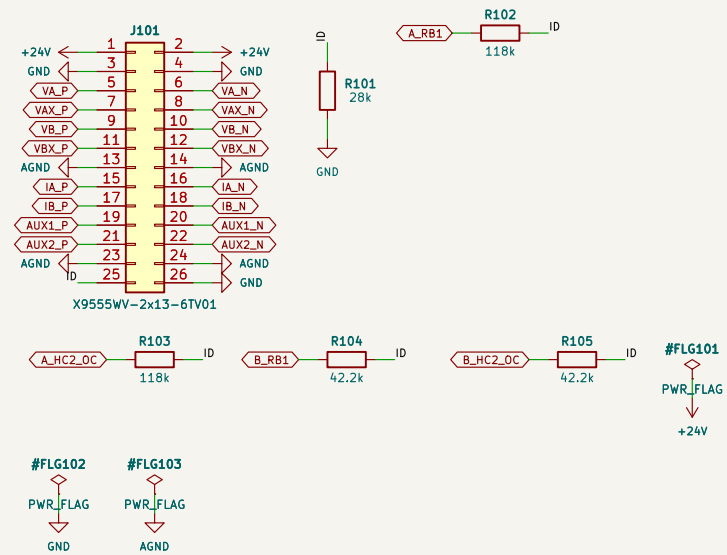
IC–17/IC–18: the whole board is conformal–coated to PD1 (IEC 60664–3 Type 1) over all HV copper and every HV–PELV isolator footprint (8–8.5 mm packages < 10 mm reinforced at PD2). Only the NH fuse links / bases and the DC terminals sit in the cooling–air path (PD3); the board, varistor network and branch fuses are in the closed PD2 compartment.

Surge credit (D–032, IC–01/02/03): the on–board varistor Y limits pole–PE and pole–pole to 3.8 kV at In 3 kA (calculated) and is monitored: reinforced is dimensioned for 6 kV (AMC3302 6.25 kV, AMC3330 7.7 kV pass on this case only). IC–04 OPEN: the G7L coil–contact rating (4000 V AC) is below reinforced – fix in rev F1 (own isolated coil supply).

SYS–IO–AUX: K_DISCH energised first at power–up; the port block itself inhibits the discharge of a closed port. Coil flyback never reaches the DO clamps in normal operation (two–switch coil drives, varistor / zener across each coil).

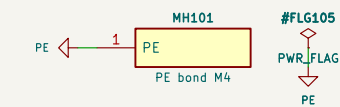
Not bench–validated. Values: sim/port_design.py → port_spec.json		
Sheet 1		
Schematics authored by Chinmoy Bhuyan		
HV–PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV–PE basic + bonding		
Sheet: /		
File: PV–PORT.kicad_sch		
Title: PV–PORT 135 A port board – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 1/7

PORT CONNECTOR (CTRL-C2000) + ID WITH HOLD READBACK



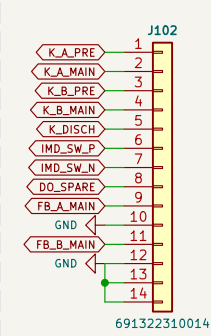
interfaces.PORT, ribbon 1:1. AMC outputs drive the cable directly (≤ 500 pF, AMC3330 p4); anti-alias RC is on CTRL-C2000. ID: 28k || readback network = 9.99 k (PV=PORT code 10.0k; CTRL-C2000 rev D values). RB outputs through 2 x 118k (port A) and 2 x 42.2k (port B): nine states, ≥ 155 counts apart (81 with the hold coils on, sim/port_design.py sec. 7.4); CTRL decodes them with its port-only table.

PE BOND



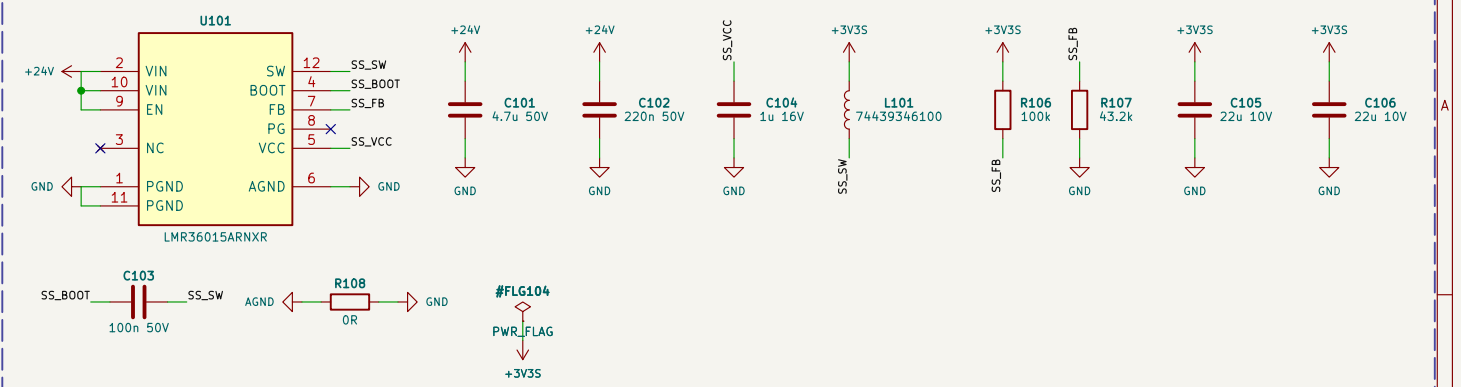
Board PE to chassis at the SPD / Y-capacitor end.

CONTROL CONNECTOR (SYS-IO-AUX)



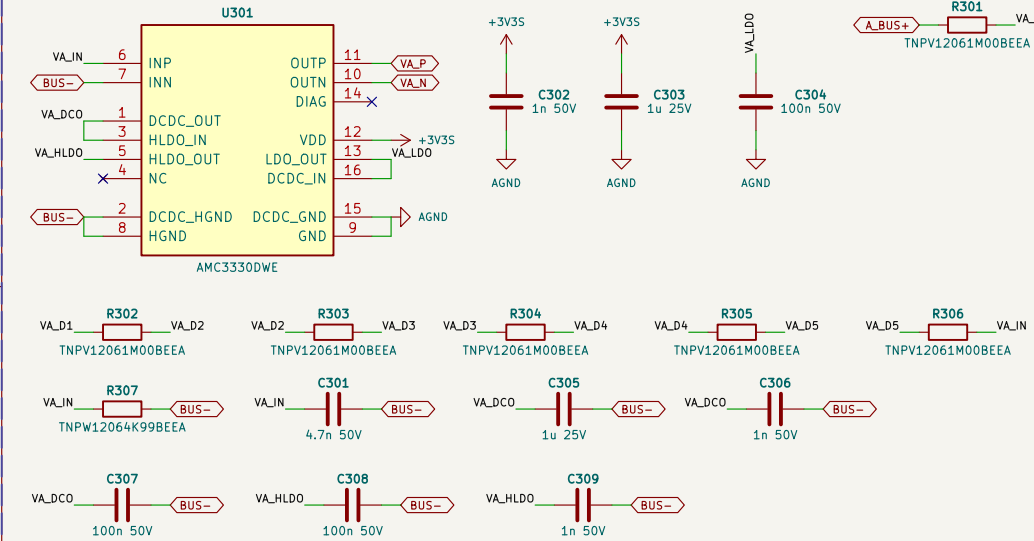
port.CTRL_PINS: DO1-DO8 (DO_SPARE = proof test), FB_A + its GND, FB_B + its GND (INT-02), 2 coil returns. Port B (battery) drives 2 HVC43 coils: K_B_MAIN up to 0.60 A -> SYS DO4 needs ILIM 10 k (1.58-2.30 A).

SENSOR SUPPLY +3V3S



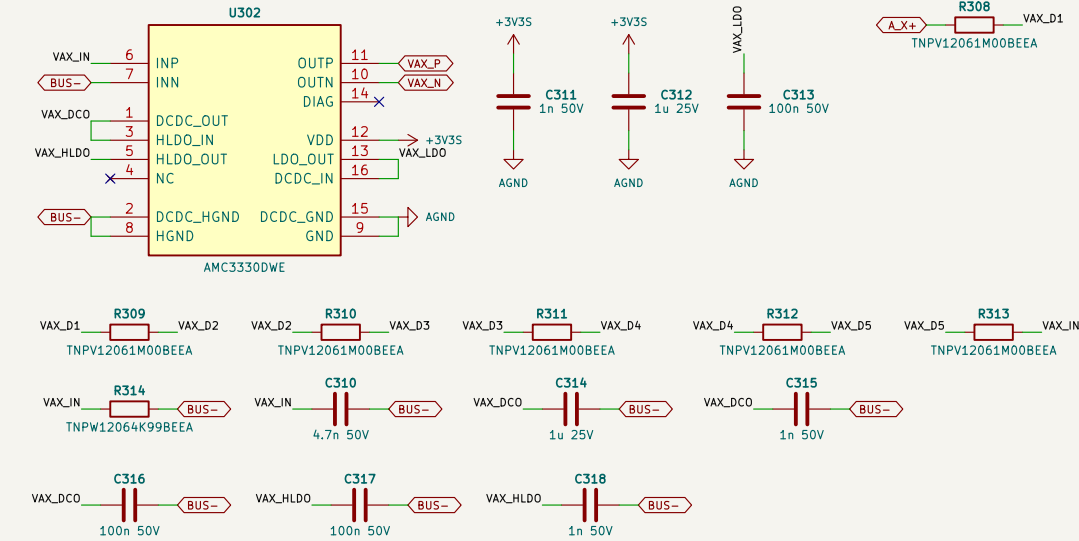
LMR36015 400 kHz, 24 V -> 3.315 V (100 k / 43.2 k, SNVSB49D sec 10.2), <= 0.34 A for 6 x AMC3330 + 2 x AMC3302 + hold logic. AGND joins GND only here (0 ohm).

VA – BUS-SIDE VOLTAGE



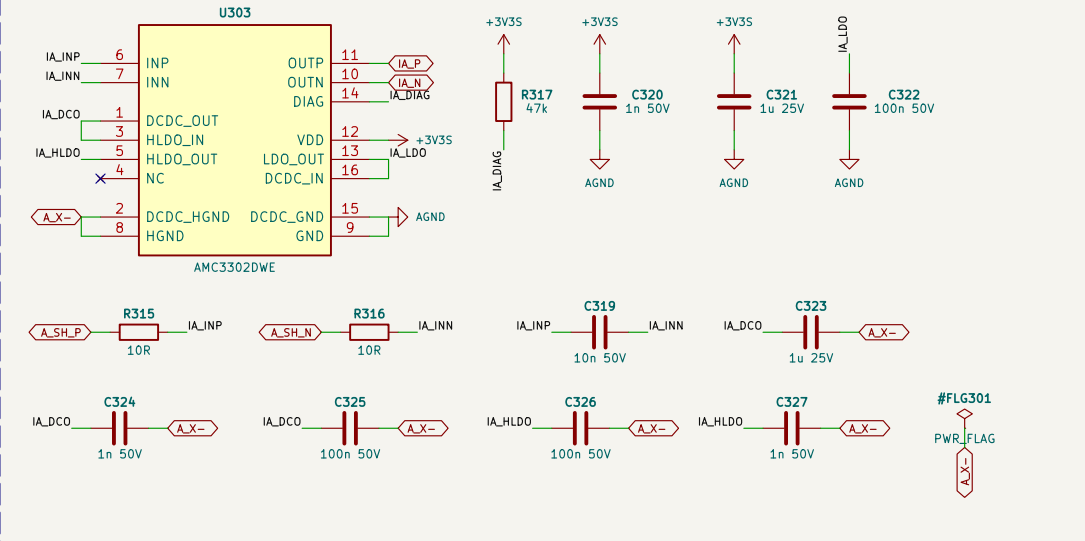
6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF: lag
 <= 27 us incl. AMC3330. Out: VA_P/N (+/-2 V on 1.44 V, PORT contract). HGND = BUS-.

VAX – TERMINAL-SIDE VOLTAGE



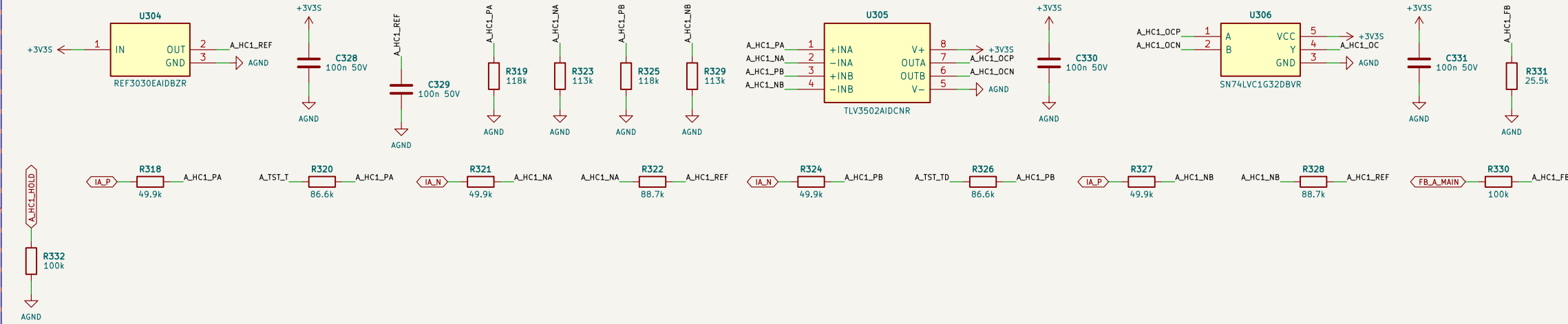
6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF: lag
 <= 27 us incl. AMC3330. Out: VAX_P/N (+/-2 V on 1.44 V, PORT contract). HGND = BUS-.

IA – PORT CURRENT



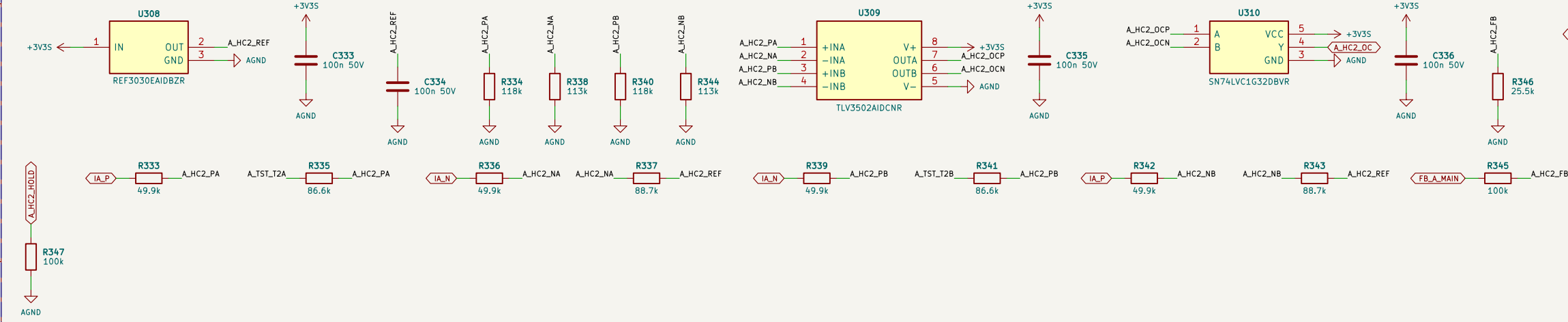
Linear ± 500 A, clip ± 640 A (OC trip 202 A, SC trip 405 A inside), 10 ohm + 10 nF input filter (SBASA11B Fig 8-2), HGND on the shunt's X- bolt by its own trace. Fail-safe out = -2.57 V.

HOLD-CLOSED LOGIC, CHANNEL 1 (IRFR9214 HIGH SIDE)



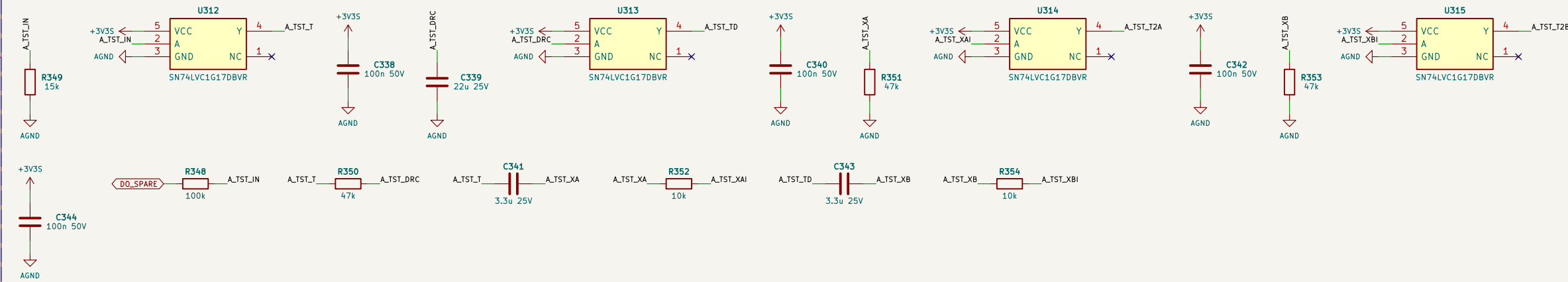
|IA| > 410 A (band 382-441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND DIAG = valid -> HOLD1. Test injection 86.6 k from A_TST_I (side A) / A_TST_TD (side B): shifts the threshold by 459 A, so a healthy side trips at 0 A under TEST only.

HOLD-CLOSED LOGIC, CHANNEL 2 (IRFL214 LOW SIDE)



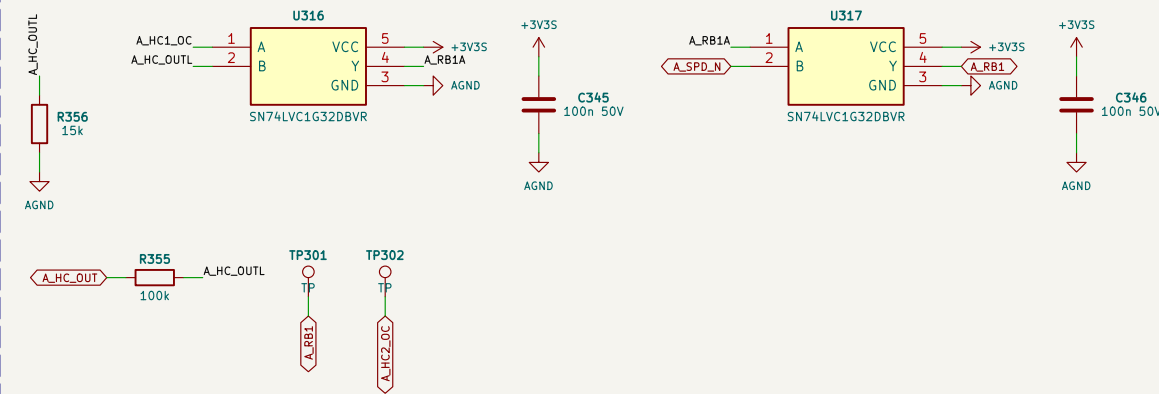
|IA| > 410 A (band 382-441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND DIAG = valid -> HOLD2. Test injection 86.6 k from A_TST_T2A (side A) / A_TST_T2B (side B): shifts the threshold by 459 A, so a healthy side trips at 0 A under TEST only.

PROOF-TEST INJECTION (DO_SPARE = TEST)



100 k / 15 k $\rightarrow$ Schmitt buffer T (ch1 side A). T $\rightarrow$ 47 k / 22 μ F $\rightarrow$ T_d (ch1 side B). One-shots (3.3 μ / 47 k) of T $\rightarrow$ T2A (ch2 side A) and of T_d $\rightarrow$ T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, contactor open and closed (report sec. 7.4: windows, what it finds and what it does not).

READBACK RB1 / RB2 (TO CTRL ON A SPARE LINE)



RB1 = OC1 OR hold high-side output present (100 k / 15 k); at rest RB1 = 1 means a stuck channel 1.
RB2 = OC2. PV-PORT sums them onto the ID line (board_io); firmware reads RB1 + RB2 against the measured current in operation and through the proof-test windows at start.

Not bench-validated. Values: sim/port_design.py -> port_spec.json
Sheet 4
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1): HV-PE basic + bonding

Sheet: /03 Port A sensing and hold logic/
File: 03_portA_sense.kicad_sch

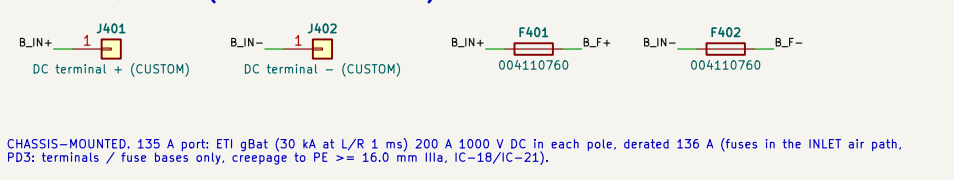
Title: PV-PORT 135 A port board – Port A sensing and hold logic

Size: A2	Date: 2026-10-04	Rev: F0
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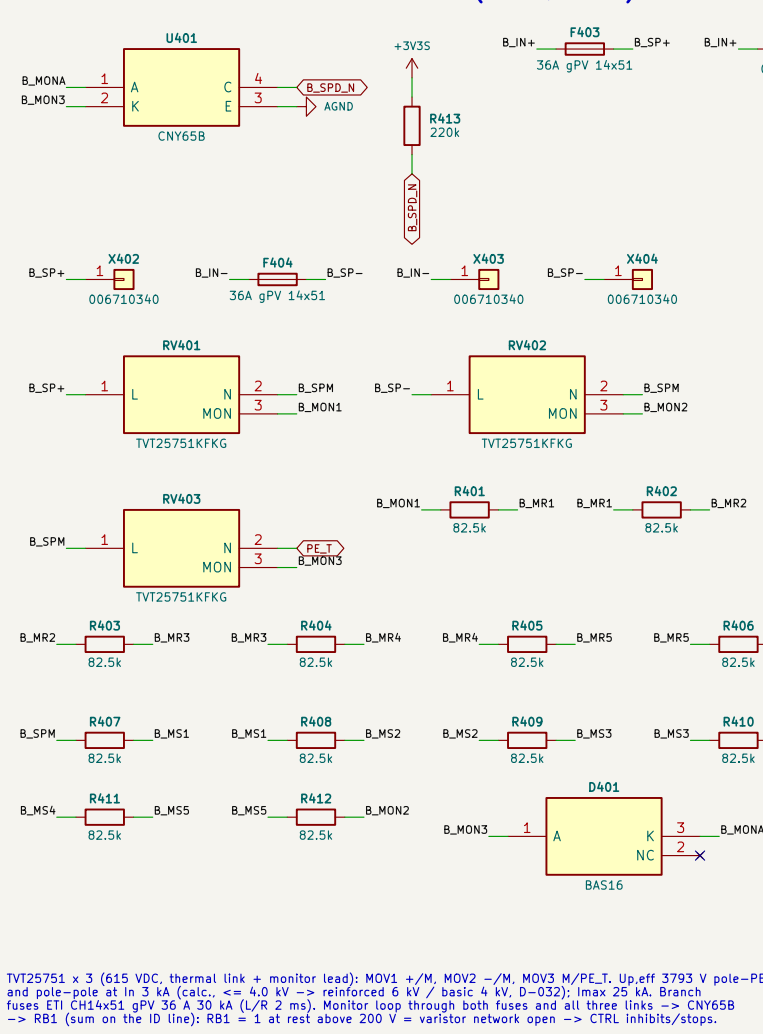
KiCad E.D.A. 10.0.6

Rev: F0

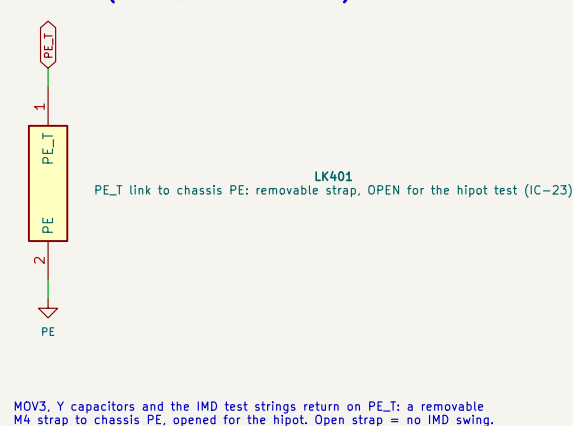
TERMINALS, FUSES (BATTERY-TYPE PORT)



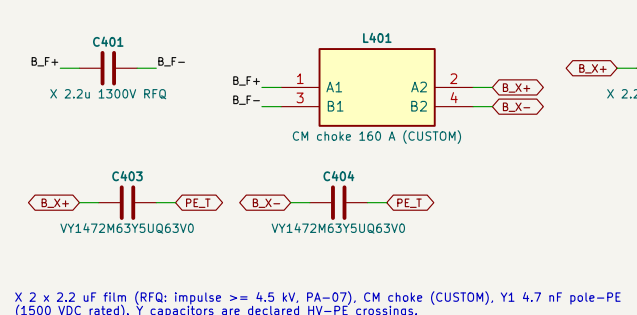
ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



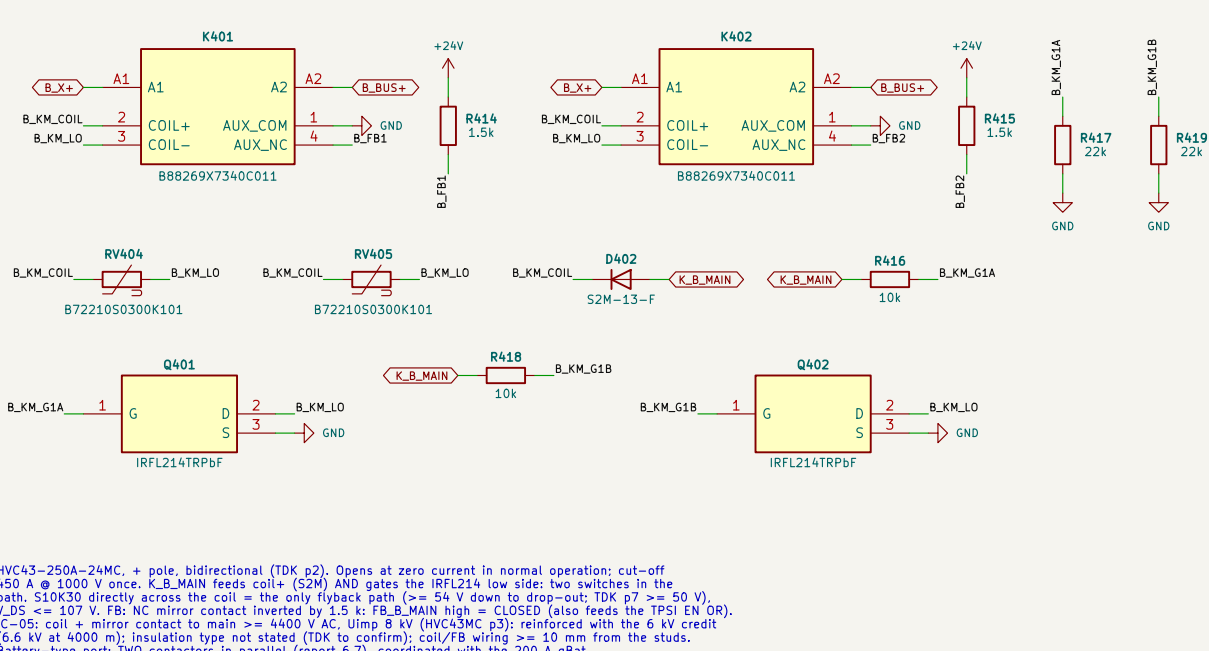
PE LINK (IC-23, IMD LESSON 4)



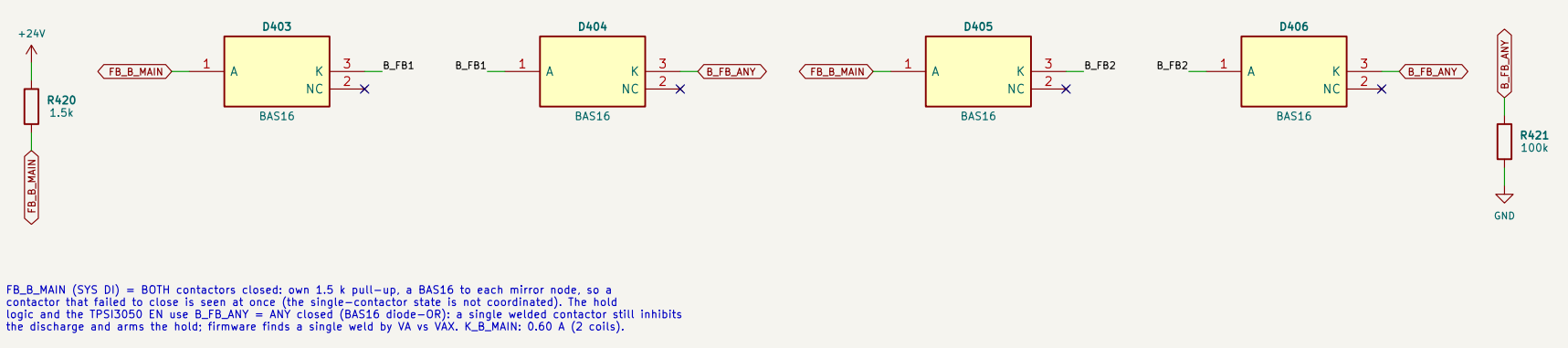
EMI FILTER



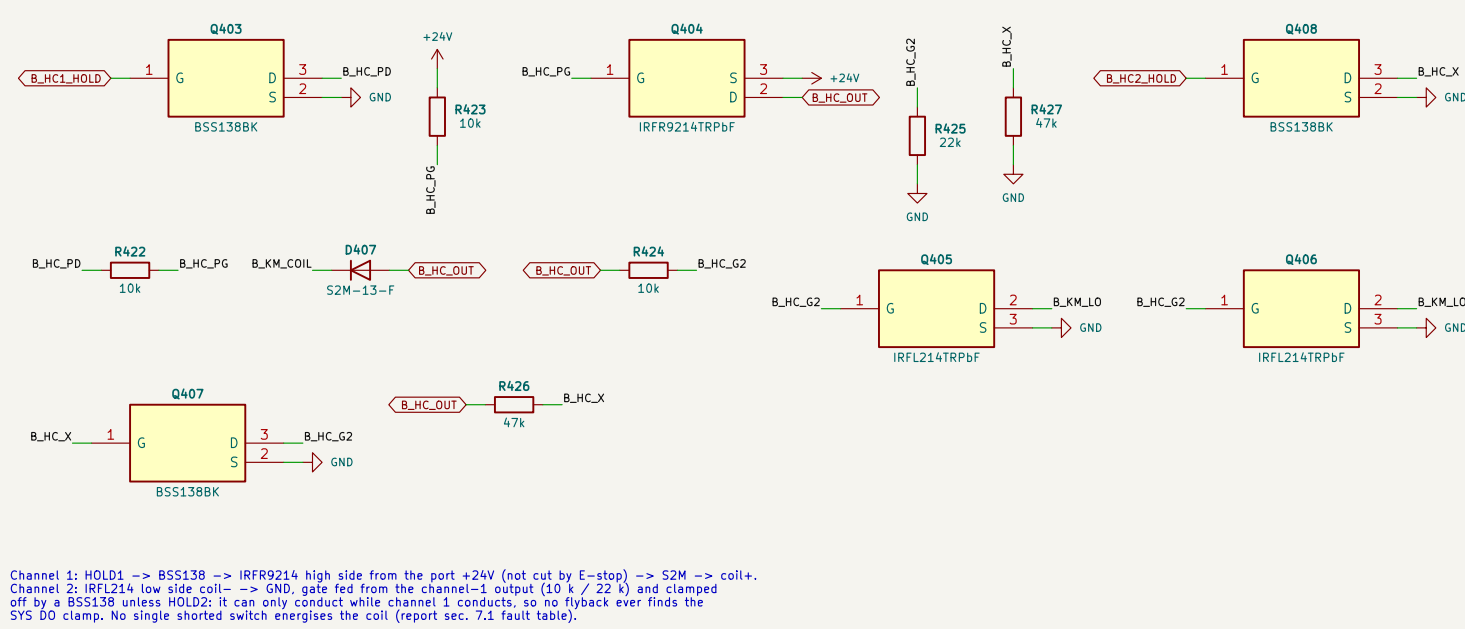
MAIN CONTACTOR: COIL DRIVE + FEEDBACK



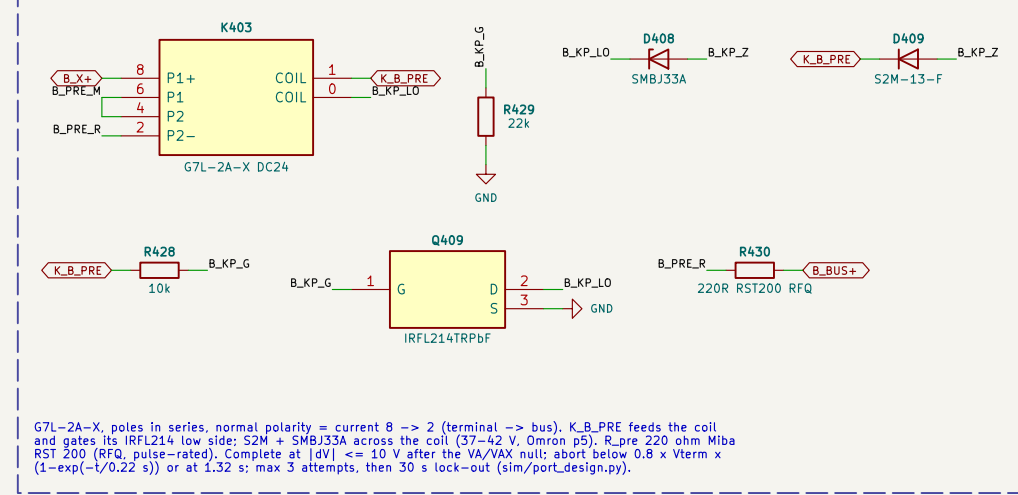
CONTACTOR PAIR FEEDBACK (BATTERY-TYPE PORT)



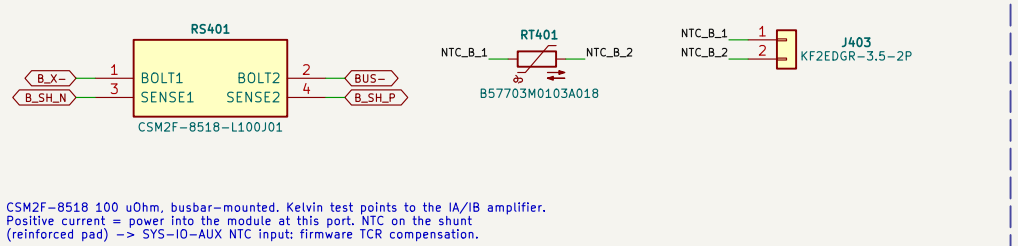
HOLD-CLOSED POWER STAGE (2 INDEPENDENT CHANNELS)



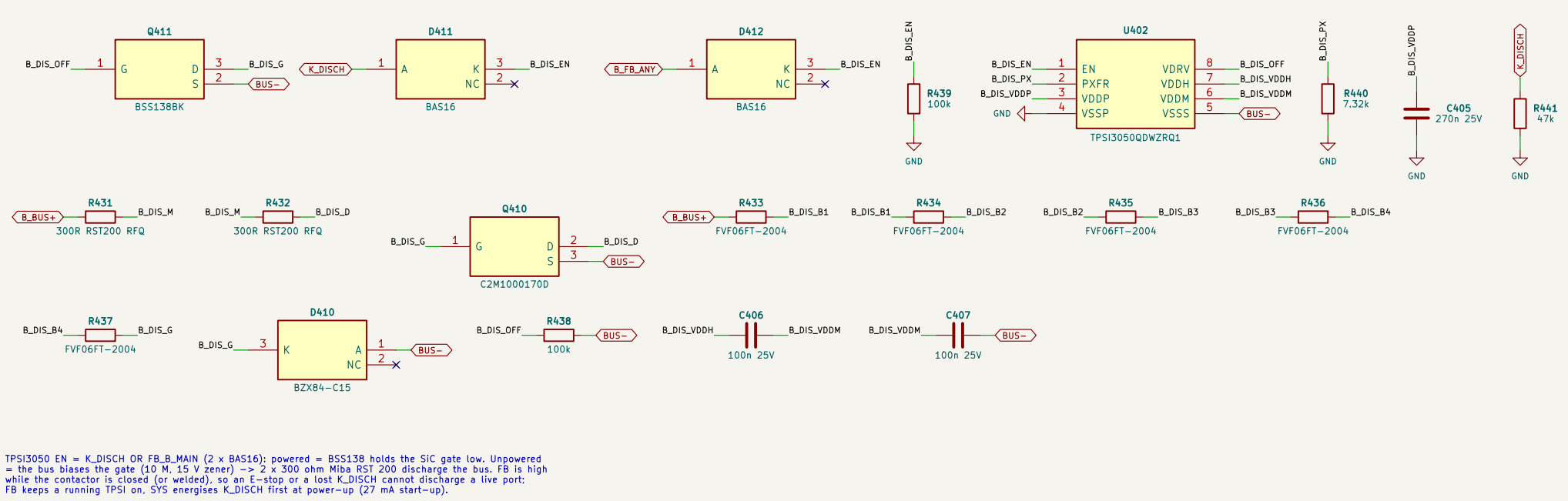
PRECHARGE RELAY + RESISTOR



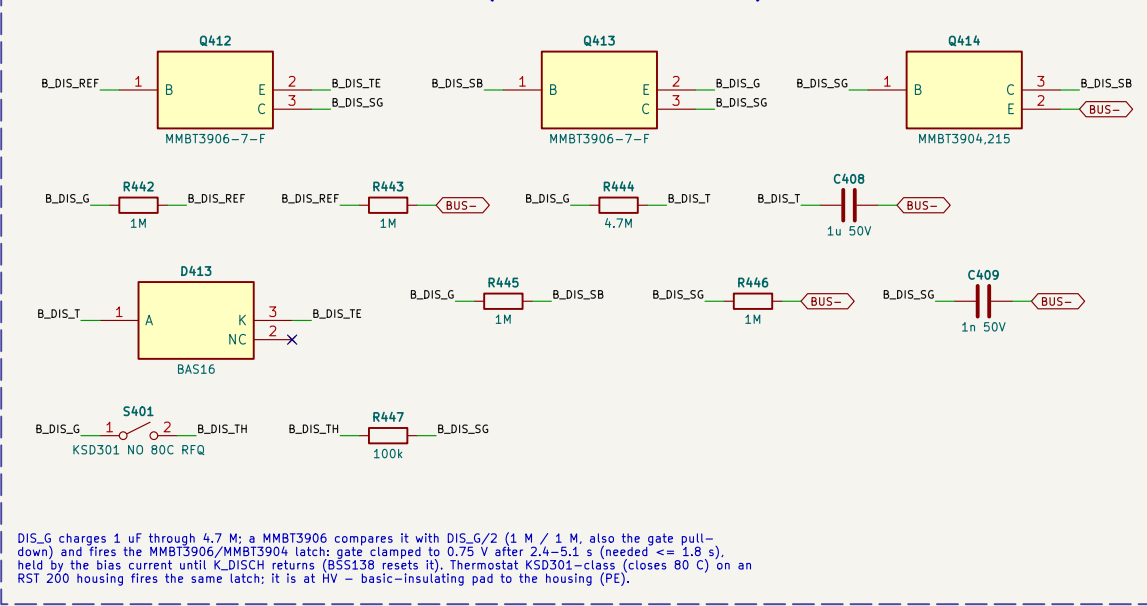
SHUNT (– POLE) + SHUNT NTC



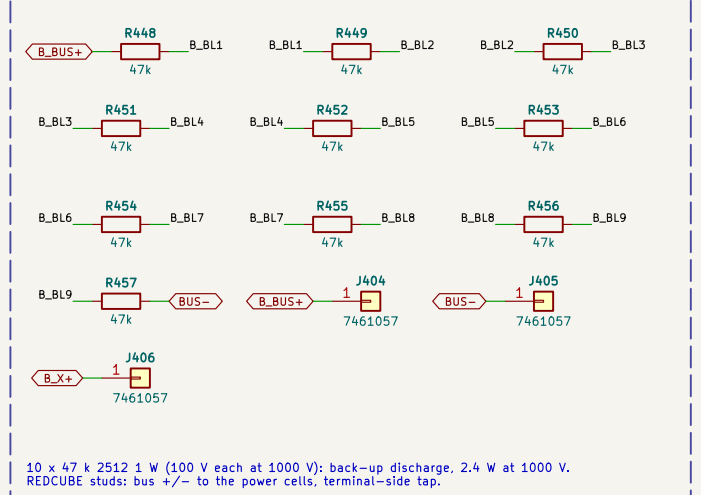
FAIL-SAFE ACTIVE DISCHARGE, INHIBITED WHILE THE CONTACTOR IS CLOSED



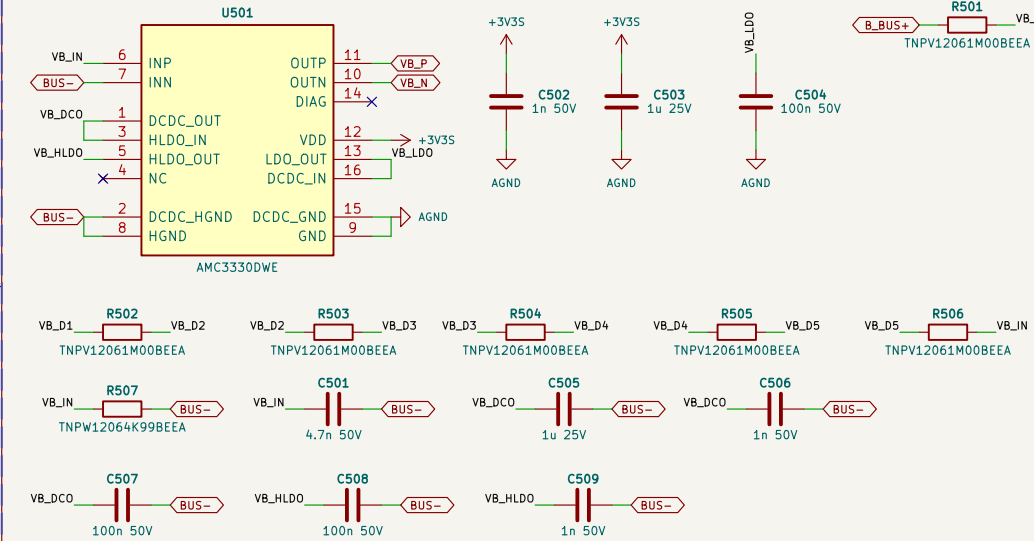
DISCHARGE TIME LIMIT + THERMAL CUT-OFF (HV SIDE, NEEDS NO 24 V)



PASSIVE BLEEDER + BOARD CONNECTION POINTS

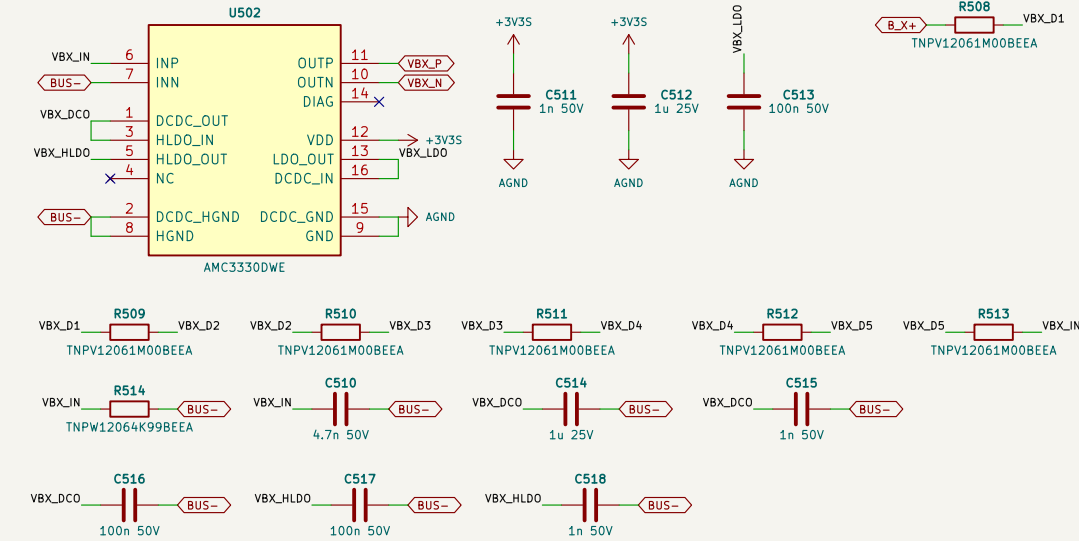


VB – BUS-SIDE VOLTAGE



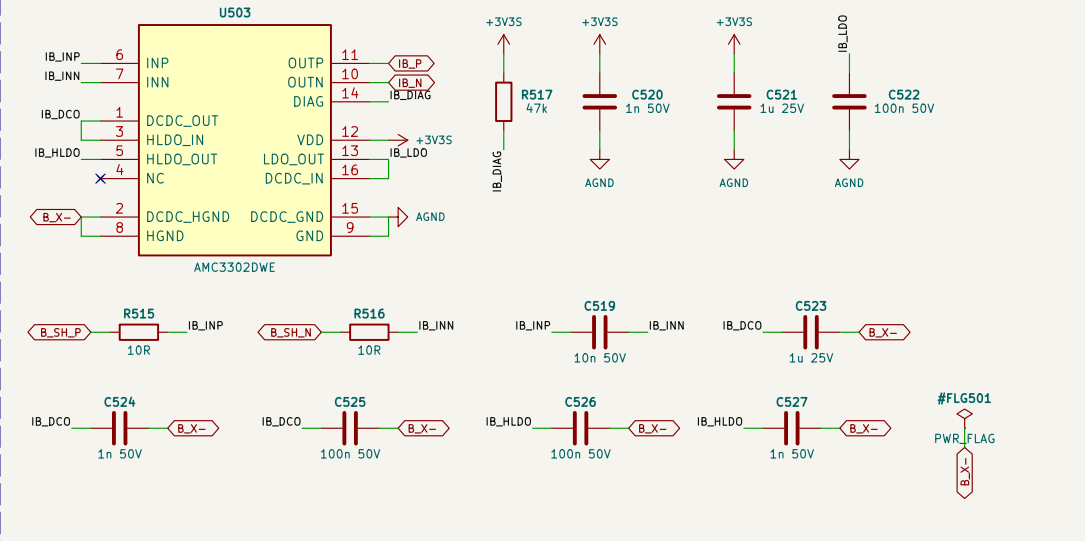
6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF: lag <= 27 us incl. AMC3330. Out: VB_P/N (+/-2 V on 1.44 V, PORT contract). HGND = BUS-.

VBX – TERMINAL-SIDE VOLTAGE



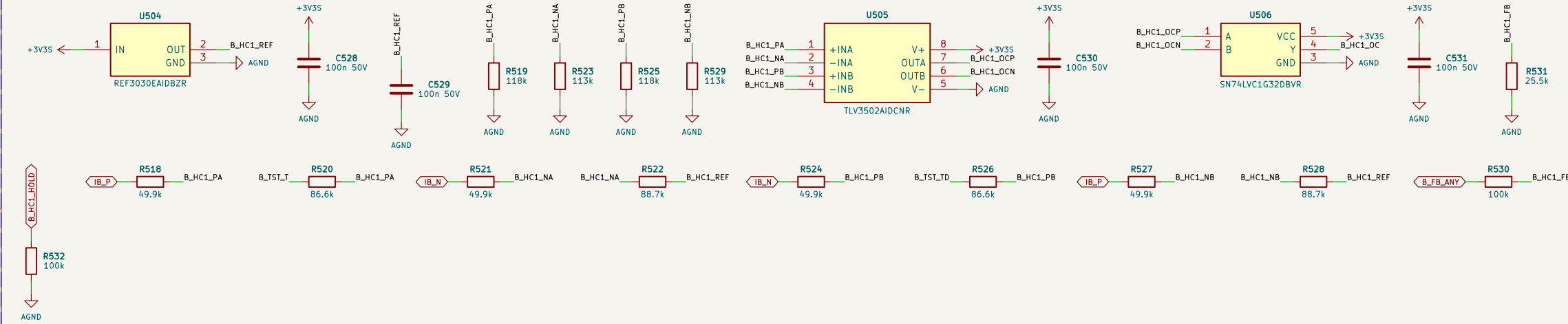
6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF: lag
 <= 27 us incl. AMC3330. Out: VBX_P/N (+/- 2 V on 1.44 V, PORT contract). HGND = BUS-.

IB – PORT CURRENT



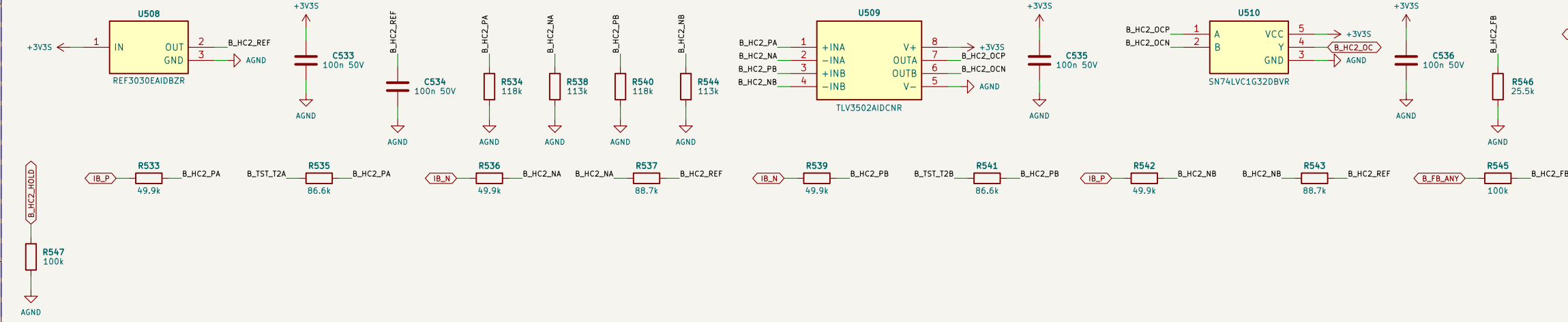
Linear ± 500 A, clip ± 640 A (OC trip 202 A, SC trip 405 A inside). 10 ohm + 10 nF input filter (SBASA11B Fig 8-2). HGND on the shunt's X- bolt by its own trace. Fail-safe out = -2.57 V

HOLD-CLOSED LOGIC, CHANNEL 1 (IRFR9214 HIGH SIDE)



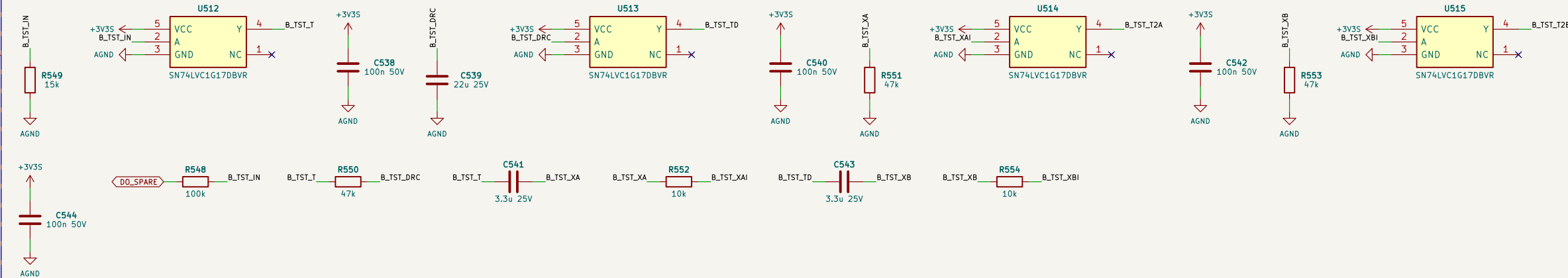
|IB| > 410 A (band 382-441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND DIAG = valid -> HOLD1. Test injection 86.6 k from B_TST_T (side A) / B_TST_TD (side B): shifts the threshold by 459 A, so a healthy side trips at 0 A under TEST only.

HOLD-CLOSED LOGIC, CHANNEL 2 (IRFL214 LOW SIDE)



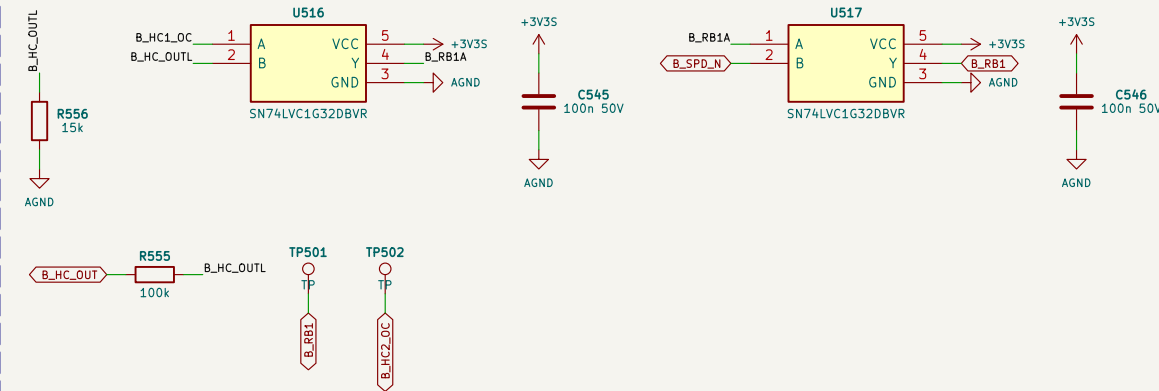
|IB| > 410 A (band 382-441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND DIAG = valid -> HOLD2. Test injection 86.6 k from B_TST_T2A (side A) / B_TST_T2B (side B): shifts the threshold by 459 A, so a healthy side trips at 0 A under TEST only.

PROOF-TEST INJECTION (DO_SPARE = TEST)



100 k / 15 k $\rightarrow$ Schmitt buffer T (ch1 side A). T $\rightarrow$ 47 k / 22 μ F $\rightarrow$ T_d (ch1 side B). One-shot (3.3 μ s / 47 k) of T $\rightarrow$ T2A (ch2 side A) and of T_d $\rightarrow$ T2B (ch2 side B). No single fault trips both channels for longer than one-shot delay. Firmware runs it at every start, contactor open and closed (report sec. 7.4: windows, what it finds and what it does not).

READBACK RB1 / RB2 (TO CTRL ON A SPARE LINE)



RB1 = OC1 OR hold high-side output present (100 k / 15 k); at rest RB1 = 1 means a stuck channel 1.
RB2 = OC2. PV-PORT sums them onto the ID line (board_io); firmware reads RB1 + RB2 against the measured current in operation and through the proof-test windows at start.

Not bench-validated. Values: sim/port_design.py -> port_spec.json
Sheet 6
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1): HV-PE basic + bonding

Sheet: /05 Port B sensing and hold logic/
File: 05_portB_sense.kicad_sch

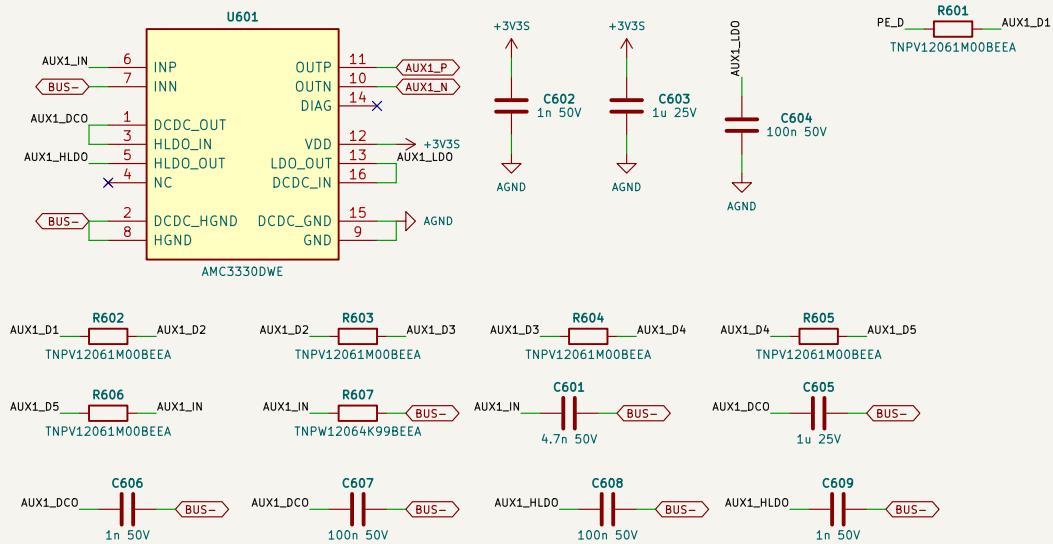
Title: PV-PORT 135 A port board – Port B sensing and hold logic

Size: A2	Date: 2026-10-04	Rev: F0
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KiCad E.D.A. 10.0.6

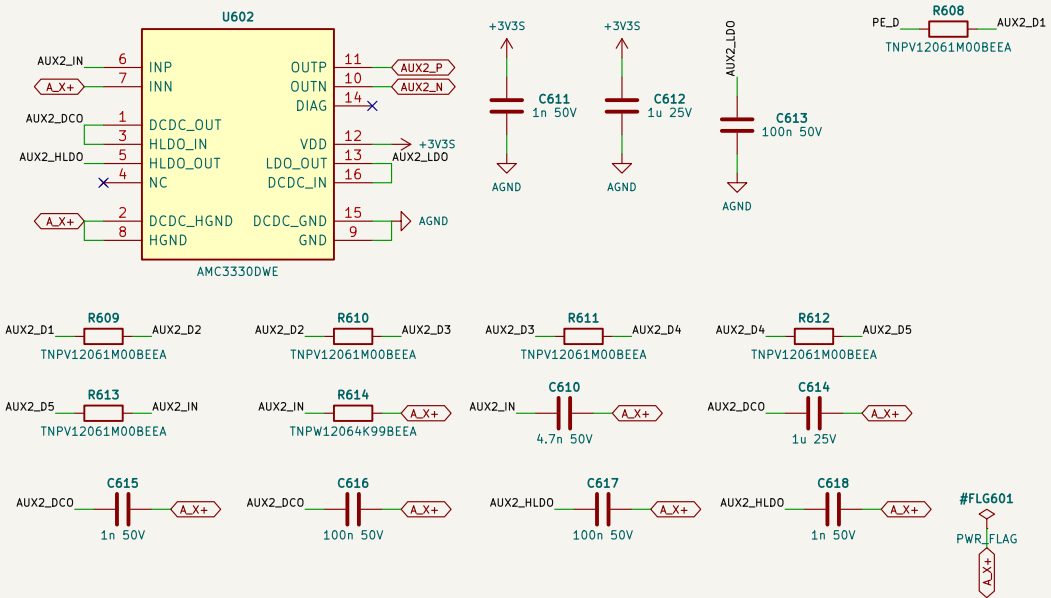
Rev: F0

AUX1 – PE RELATIVE TO BUS–



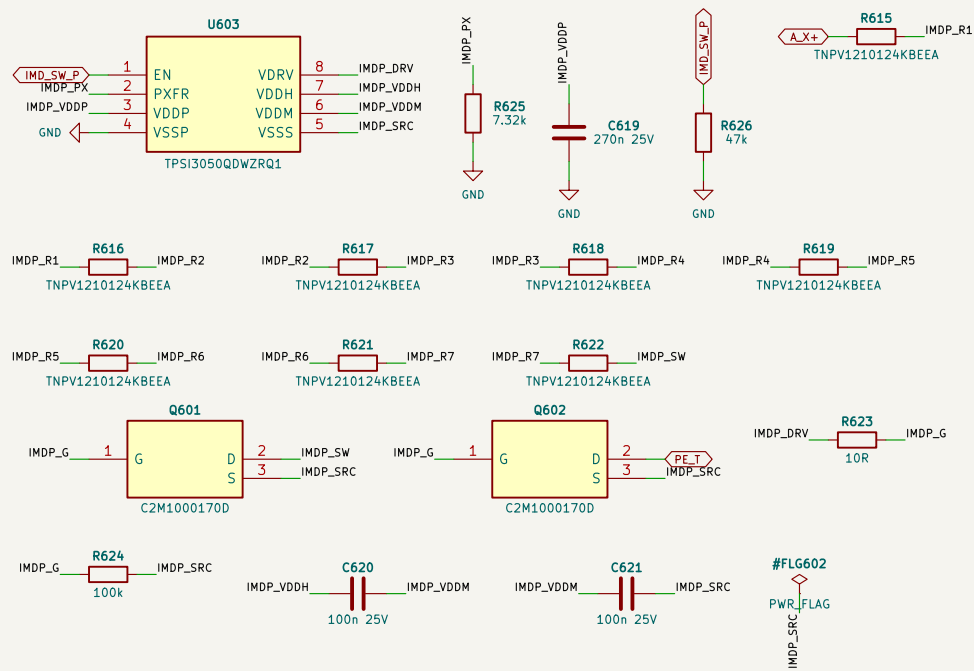
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

AUX2 – PE RELATIVE TO A_X+



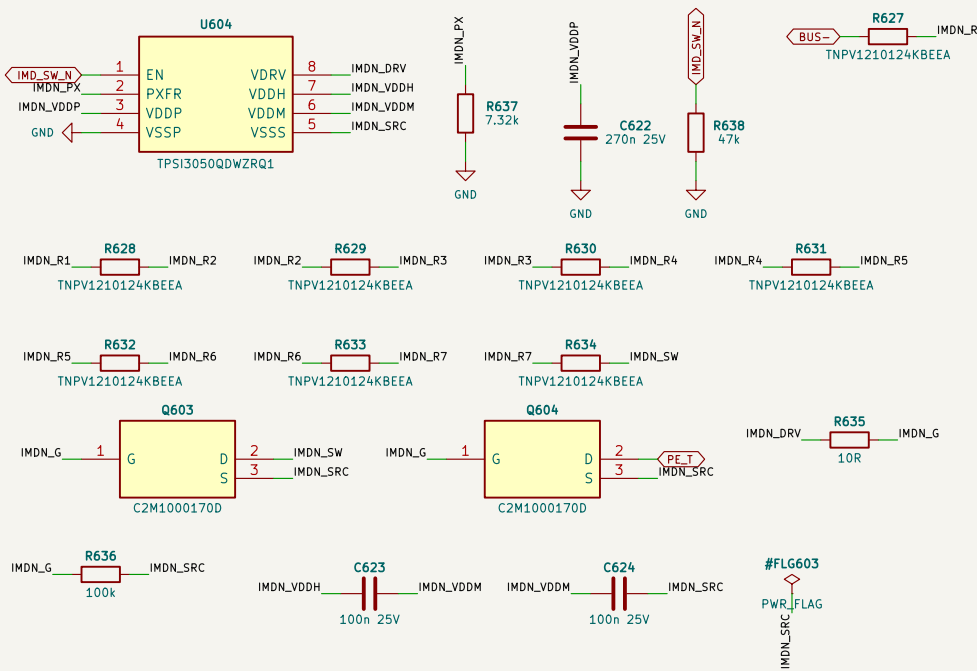
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

TEST STRING A_X+ -> PE (IMD_SW_P)



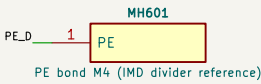
8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

TEST STRING BUS– -> PE (IMD_SW_N)



8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

PE_D BOND + ONE ACTIVE MONITOR PER DC SYSTEM



PE_D: own M4 bond (dividers only). The module disables its own measurement by leaving IMD_SW_P/N de-energised (strings open, fail-safe): a passive module still loads each pole with 6.005 M to PE.

Not bench-validated. Values: sim/port_design.py -> port_spec.json
Sheet 7
Schematics authored by Chinmoy Bhuyan
HV–PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV–PE basic + bonding

Sheet: /06 Insulation measurement/
File: 06_imd.kicad_sch

Title: PV–PORT 135 A port board – Insulation measurement

Size: A3 Date: 2026–10–04 Rev: F0
KiCad E.D.A. 10.0.6 Id: 7/7